Project Log: Benchmark Noise Thresholds

Per-benchmark minimum detectable effect (MDE) and significance guidance for the five benchmarks reported in Chen et al. (2025), *Bring Reason to Vision*: MathVista, MathVerse, MathVision, DynaMath, MMStar. Generated for the session/step_weight_merging worktree on 2026-05-31.

1. Why this matters

Reported accuracy changes between a base VLM and a weight-merged variant are only meaningful if they exceed the test set's intrinsic sampling noise. A naive +0.4 pp gain on a 200-sample slice can easily be sampling variance, whereas the same gain on a 5,000-sample slice is much more likely real. This document collects per-benchmark sample sizes and the resulting noise floor so that future comparisons can be screened consistently.

2. Methodology

Accuracy is a proportion p measured over n independent test items. The Wald 95% confidence-interval half-width on p is:

$$\mathrm{MDE}_{\mathrm{unpaired}} = 1.96 \times \sqrt{\left(p(1-p) / n \right)}$$

This is the conservative noise floor for two independently sampled accuracy estimates. For the typical case where the *same* test items are scored under both conditions (baseline vs. merged model), the paired McNemar test is tighter:

$$\mathrm{MDE}_{\mathrm{paired}} \approx 1.96 \times \sqrt{\left((b+c) / n^2 \right)}$$

where $b+c$ is the number of items on which the two models disagree. For VLM merging comparisons we typically observe $(b+c)/n \approx 0.10$–0.20, giving a paired MDE roughly **1.5–2× tighter** than the unpaired Wald bound. The paper's own significance test (Appendix G, Table 9) implicitly uses the paired version, which is why Δ=+1.2 pp on MathVista-Math is marked $p<0.05$ even though the unpaired Wald MDE is ~3.9 pp.

Practical rule of thumb used below: we report the Wald MDE as a conservative ceiling and a paired-test estimate as the realistic threshold. A reported change should clear the paired threshold before being treated as non-noise; changes between the paired and Wald thresholds are *plausibly* real but warrant a McNemar test before reporting.

3. Per-benchmark thresholds

Benchmark	n	Typical p	Wald MDE (pp)	Paired MDE est. (pp)	Notes
MathVista-All (testmini)	1000	0.37-0.66	2.9-3.1	~1.5-2.0	Full testmini. Per MathVista paper, testmini distribution closely ma
MathVista-General (testmini)	460	0.50-0.67	4.3-4.7	~2.3-2.5	General-VQA subset (TSV category=general-vqa, 460 rows). Falls
MathVista-Math (testmini)	540	0.25-0.65	3.4-4.1	~1.8-2.2	Math-targeted subset (TSV category=math-targeted-vqa, 540 rows
MathVerse (per mode)	788	0.11-0.54	2.2-3.5	~1.2-1.8	Each of T-D, T-L, V-I, V-D, V-O has 788 problems. Noise floor sca
MathVerse-Overall	788 (effective)	0.19-0.44	2.8-3.5	~1.5-1.8	Avg of 5 modes on same 788 problems => effective n stays ~788
MathVision_MINI (testmini)	304	0.14-0.25	3.9-4.9	~2.0-2.6	Standard testmini subset used by BRV (dataset name MathVision_
DynaMath (Average)	5010	0.22-0.39	1.1-1.4	~0.5-0.7	10 variants/seed; nominal Wald assumes independence. BRV pap

Benchmark	n	Typical p	Wald MDE (pp)	Paired MDE est. (pp)	Notes
DynaMath (Worst Case)	501	0.10-0.35	2.6-4.2	~1.4-2.2	% of seeds correct on ALL 10 variants. Already produced by VLME
MMStar-All	1500	0.44-0.67	2.4-2.5	~1.3-1.4	6 capability dimensions x ~250 items each.
MMStar-Math (per dimension)	~250	0.30-0.75	5.4-6.0	~2.8-3.2	Single 250-item axis. The largest floor in this table; treat <3 pp cha

How to read: a Δ below the *paired* column is almost certainly noise. A Δ between paired and Wald is suggestive; run a McNemar test before claiming it. A Δ above Wald is robust under either testing regime.

4. Sanity-check against the paper's own significance test

Chen et al. (2025) Appendix G marks $p<0.05$ cells with *. Comparing those marks to our paired-MDE estimates (LLaVA-Next-LLaMA3-8B + Dart-Uniform):

Benchmark	Reported Δ	Paper sig?	Our paired MDE est.	Consistent?
MathVista-Math	+2.9	yes (*)	~1.8-2.2 pp	yes
MathVerse-Overall	+3.5	yes (*)	~1.5-1.8 pp	yes
MathVerse-T-D	+6.1	yes (*)	~1.5-1.8 pp	yes
MathVerse-T-L	+4.8	yes (*)	~1.5-1.8 pp	yes
MathVerse-V-I	+4.3	yes (*)	~1.5-1.8 pp	yes
MathVerse-V-D	+2.8	yes (*)	~1.5-1.8 pp	yes
MathVerse-V-O	+1.4	no	~1.2-1.4 pp	borderline (paper says not sig; our est. says margina
MMStar-Math	+1.2	no	~2.8-3.2 pp	yes (sub-threshold)
DynaMath	+1.8	no	~0.5-0.7 pp (or ~1.4-2.2 worst-case)	borderline (depends on avg vs. worst-case n)
MathVista-Math (+MAmmoTH-2)	+0.3	no	~1.8-2.2 pp	yes (sub-threshold)
MathVista-Math (+Magpie)	+0.5	no	~1.8-2.2 pp	yes (sub-threshold)

Our paired-MDE estimates agree with the paper's *-marks in 10/11 cases. The single disagreement (MathVerse-V-O +1.4) is borderline: V-O is the smallest-p subset (~0.16), where the paired MDE drops to ~1.2-1.4 pp. A 1.4 pp gain sits right on the boundary, consistent with the paper not marking it significant.

5. Recommendations for the weight-merging pipeline

When evaluating step 13 (text_inject) / step 25 (weight_merge / P5) outputs against the baseline VLM:

- Treat any per-benchmark Δ below the paired MDE estimate in the table as **noise**.
- For Δ in the gray zone (paired < Δ < Wald), run a paired McNemar test on the per-question correctness flags before reporting.
- For sub-axes with $n < 300$ (MMStar-Math, MathVision-Wild, DynaMath worst-case), require >3 pp before treating as a real change.
- For aggregate scores like MathVerse-Overall, do NOT use the 3,940 question count for the Wald formula -- the five modes share the same 788 problems, so the effective n is closer to 788.
- For DynaMath, the score CSV already contains both Setting=Average (n=5,010, correlated within seed) and Setting=Worst Case (n=501, independent). Apply each row's MDE separately; do not compare an Average Δ against a Worst-Case threshold or vice versa.

6. Why qwen25vl-3b cannot be weight-merged (no math donor available)

Linear weight merging via task arithmetic requires the donor LM to have **parameter-by-parameter shape match** with the VLM's language-model component (Ilharco et al. 2023; Chen et al. 2025 §2). Concretely, $\theta'\_vlm = \alpha \cdot \theta\_vlm + (1-\alpha) \cdot \theta\_donor$ is an element-wise tensor op — every tensor in θ_donor must have the same shape as the corresponding tensor in θ_vlm. Different parameter counts cannot be reconciled without retraining.

Qwen2.5-VL-3B's language model is Qwen2.5-3B (architecture: Qwen2; hidden_size=2048, intermediate_size=11008, num_layers=36 from the --model-type qwen25vl-3b defaults in run_pipeline.sh). A math-fine-tuned donor of the same shape would have to be a Qwen2.5-3B checkpoint trained on math.

Available Qwen2.5-Math sizes (Hugging Face + ModelScope, the official Alibaba release):

- **Qwen2.5-Math-1.5B** — shape mismatch (1.5B params vs 3B)
- **Qwen2.5-Math-7B** — shape mismatch (7B params vs 3B)
- **Qwen2.5-Math-72B** — shape mismatch
- **no Qwen2.5-Math-3B** — confirmed by the Qwen2.5-Math technical report (arXiv:2409.12122) and the QwenLM/Qwen2.5-Math GitHub repository, which list only 1.5B, 7B, 72B variants

Off-Hugging-Face search results (also ModelScope, GitHub, ArXiv):

- **QwenLM/Qwen2.5-Math** (Alibaba GitHub) — same release as HF, no 3B variant.
- **ModelScope (Alibaba's hub)** — mirrors the HF release; no 3B Qwen2.5-Math.
- **DeepScaleR-1.5B-Preview** (Agentica) — built on DeepSeek-R1-Distill-Qwen-1.5B, not parameter-compatible with Qwen2.5-3B.
- **Qwen-2.5-Math-7B-SimpleRL** (hkust-nlp) — 7B, not 3B.
- **Open-R1** — trains 7B / 8B / 14B / 32B variants on Qwen / Llama bases; no 3B Qwen2.5 checkpoint.
- **Community LoRA fine-tunes** on Qwen2.5-3B for GSM8K (e.g., ashkunwar/COT_Finetuning) — these ship LoRA *adapters*, not full-shape checkpoints. To use one for task arithmetic you'd have to materialize the adapter into a full delta (LoRA merge), then take the delta against base — doable but defeats the “free-lunch” promise of BRV-style merging.

Implication for our experiment matrix:

qwen25vl-3b is excluded from the baseline-vs-uniform-λ=0.9 batch. Of the 9 model-types run_pipeline.sh supports for step 25, the 8 remaining have parameter-compatible math donors:

--model-type	VLM size	Math donor	Source	Status
llava-hf	7B	MetaMath-7B-V1.0	HF	todo
llava-mistral	7B	MAmmoTH-7B-Mistral	local	todo
llava-llama3	8B	dart-math-llama3-8b-prop2diff	local	done
llava-ov	7B	Qwen2-Math-7B	HF	todo
qwen2vl	7B	Qwen2-Math-7B	HF	done (re-merged)
qwen25vl-7b	7B	Qwen2.5-Math-7B	HF	todo
qwen25vl-3b	3B	— (no compatible donor)	—	blocked
idefics2	8B	MAmmoTH-7B-Mistral	local	done
internvl	8B	internlm2_5-math-7b	HF	todo

If you do want a qwen25vl-3b row eventually: the cheapest path is to take a community Qwen2.5-3B math LoRA (e.g., from GitHub COT_Finetuning), merge the LoRA into a full Qwen2.5-3B checkpoint via PEFT's merge_and_unload(), and use that as the donor. Caveat: a LoRA-derived donor is a much weaker “task vector” than Alibaba's official Qwen2.5-Math-7B (which the larger qwen25vl-7b row uses) — expect smaller gains, and the comparison to BRV would no longer be apples-to-apples.

7. References

- Chen, S. et al. (2025). *Bring Reason to Vision: Understanding Perception and Reasoning through Model Merging*. ICML 2025. arXiv:2505.05464v2. https://arxiv.org/abs/2505.05464 — Appendix G describes their paired $p<0.05$ significance test (no test type specified; consistent with McNemar).

- Lu, P. et al. (2024). *MathVista: Evaluating Mathematical Reasoning of Foundation Models in Visual Contexts*. ICLR 2024. arXiv:2310.02255. https://arxiv.org/abs/2310.02255 — testmini = 1,000 examples; KL=0.008, TV=0.035 vs. full 6,141-question set.

- Zhang, R. et al. (2025). *MathVerse: Does Your Multi-modal LLM Truly See the Diagrams in Visual Math Problems?* ECCV 2024. arXiv:2403.14624. https://arxiv.org/abs/2403.14624 — testmini = 788 problems × 5 main modes (T-D, T-L, V-I, V-D, V-O) = 3,940 question-evaluations, but 788 unique problems.

- Wang, K. et al. (2024). *Measuring Multimodal Mathematical Reasoning with MATH-Vision Dataset*. NeurIPS 2024. arXiv:2402.14804 / project: https://mathllm.github.io/mathvision/ — 3,040 full test problems; 304-sample MathVision-Wild for real-world capture robustness.

- Zou, C. et al. (2024). *DynaMath: A Dynamic Visual Benchmark for Evaluating Mathematical Reasoning Robustness of Vision Language Models*. ICLR 2025. arXiv:2411.00836. https://arxiv.org/abs/2411.00836 — 501 Python-program seeds × 10 variants = 5,010 questions. Reports avg-case A_avg, worst-case A_wst, and repetition-consistency RC (92-99% across tested models). Worst-case ~ 0.5 × avg-case across all SOTA VLMs.

- Chen, L. et al. (2024). *Are We on the Right Way for Evaluating Large Vision-Language Models?* NeurIPS 2024. arXiv:2403.20330. https://arxiv.org/abs/2403.20330 — MMStar: 1,500 vision-indispensable samples across 6 core dimensions (~250 each).

- Madaan, L. et al. (2024). *Quantifying Variance in Evaluation Benchmarks*. arXiv:2406.10229. https://arxiv.org/abs/2406.10229 — General methodology for benchmark seed/run variance; recommends accounting for variance when comparing models. Notes that for ~7B-scale models, reformulating choice tasks as completions reduces variance.

- Dietterich, T. G. (1998). *Approximate Statistical Tests for Comparing Supervised Classification Learning Algorithms*. Neural Computation 10(7). — Establishes McNemar's test as the standard for paired classifier comparison; Type I error properties.

- Raschka, S. (2018). *Model evaluation, model selection, and algorithm selection in machine learning, Part 4: McNemar & Cochran's Q tests*. https://sebastianraschka.com/blog/2018/model-evaluation-selection-part4.html — Practical McNemar implementation reference.

- Dunlap, L. et al. (2025). *Visually Prompted Benchmarks Are Surprisingly Fragile*. https://lisadunlap.github.io/vpbench/ — Empirical demonstration of the $\sqrt{N}$ dependence of benchmark CIs and how small benchmarks (100-200 samples) produce unstable VLM rankings.